

Problem Set 7: Probability and Getting Started with R

SPEA V706 Math Camp | Module 7 · Tuesday, August 11

This afternoon has two halves. **Part A** is pencil and paper, working with the probability ideas from this morning's deck. **Part B** puts R on your machine and gets you moving in it. Do Part A first; a few of the R exercises lean on answers you worked out by hand.

Reading: Hansen, *Probability and Statistics for Economists*, chapters 1–2. Decks: **M1 Probability & Random Variables**, **W2 R Basics**.

Part A. Concepts

A1. Two dice

Suppose you roll two fair six-sided dice and record the pair.

- How many outcomes are in the sample space Ω ? Are they equally likely?
- Compute $\Pr(\text{sum} = 7)$.
- Compute $\Pr(\text{at least one die shows a 6})$.
- Compute $\Pr(\text{both dice are even})$.
- Compute $\Pr(\text{sum} = 7 \mid \text{at least one die shows a 3})$. Explain in one sentence why this is not the same as your answer to (b).

A2. A cross-tabulation

A survey of 1,000 adults records employment and health insurance status.

	Insured	Not insured	Total
Employed	540	160	700
Not employed	120	180	300
Total	660	340	1,000

Treat the table as the joint distribution of two events.

- Find $\Pr(\text{Insured})$ and $\Pr(\text{Employed})$. These are the *marginal* probabilities.
- Find $\Pr(\text{Insured} \mid \text{Employed})$.

- c. Find $\Pr(\text{Employed} \mid \text{Insured})$. It is a different number than (b); say what each one means in words.
- d. Are “Insured” and “Employed” independent?

A3. A flagged tax return

An auditing system flags returns for review. In the population of returns, 4% contain a material error. The system flags 90% of returns that contain an error, but it also flags 8% of returns that do not contain an error.

- a. Write down $\Pr(E)$, $\Pr(F \mid E)$, and $\Pr(F \mid E^c)$ in the notation of the deck, where E is “contains an error” and F is “gets flagged.”
- b. Compute $\Pr(F)$, the overall probability that a return gets flagged.
- c. A return has been flagged. Compute $\Pr(E \mid F)$.
- d. Most flagged returns turn out to be clean. Explain why, in one or two sentences, referring to the base rate.

A4. Reading a CDF

Let X be the number of cars owned by a household.

x	$\Pr(X = x)$	$F(x)$
0	0.10	0.10
1	0.45	0.55
2	0.35	0.90
3	0.10	1.00

Using the CDF column wherever you can:

- a. $\Pr(X \leq 2)$
- b. $\Pr(X = 2)$
- c. $\Pr(1 < X \leq 3)$
- d. The median of X . How did you find it?

A5. From an experiment to a random variable

Suppose you flip a fair coin three times.

- a. Write out the sample space Ω . How many outcomes are there?
- b. Let X be the number of heads. For each $\omega \in \Omega$, what is $X(\omega)$?
- c. Write down the probability mass function of X .
- d. Write down the CDF $F_X(x)$ and sketch it. Where are the jumps, and how tall is each one?

Part B. R

If R and RStudio are not yet installed, do that first. Work inside an RStudio project, and keep everything you type in a script (`ps1.R`), not in the console. You should be able to re-run the whole script top to bottom.

B1. Check your setup

```
R.version.string      # which version of R are you running?
sessionInfo()         # the full picture

# pacman loads packages and installs them if they are missing
install.packages("pacman")
pacman::p_load(data.table, ggplot2, wooldridge)
```

Report your R version. If any of the three packages fails to load, say what the error was.

B2. R as a calculator, and assignment

- Compute $\frac{17+4}{3}$, then 2^{10} , then $\sqrt{144}$.
- Store the number of dice outcomes from A1(a) in an object called `n_outcomes`, then compute the probability of a single outcome as `1 / n_outcomes`.
- What is the difference between typing `x <- 5` and typing `x = 5`? What is the difference between `x <- 5` and `x == 5`?

B3. Vectors and their types

```
rolls <- c(3, 1, 6, 6, 2, 4, 5, 6)
```

- How long is `rolls`? What is its `class()`?
- Compute its mean, its maximum, and how many of the entries equal 6.
- Now run `mixed <- c(1, 2, "three")` and check `class(mixed)`. What happened to the numbers? This is the **coercion behavior** from the deck.
- Try `mean(mixed)`. Read the error message carefully and explain it.

B4. Indexing and logical subsetting

Using `rolls` from B3:

- Pull out the first element, then the last element, then elements 2 through 4.

- b. Pull out every element that is greater than 3.
- c. Explain what `rolls > 3` returns on its own, and why putting it inside square brackets does something useful.
- d. Compute the *proportion* of rolls greater than 3 with `mean(rolls > 3)`. Why does taking a mean of TRUE/FALSE values give you a proportion?

B5. Write a function

Write a function `standardize(x)` that subtracts the mean of a vector and divides by its standard deviation.

```
standardize <- function(x) {
  # your code here
}
```

- a. Test it on `rolls`. What is the mean of the result? What is its standard deviation? Explain why you should have expected those two numbers.
- b. Add a line to your function that stops with an informative message if `x` has fewer than two elements. (`stop()` is the tool.)

B6. Conditional probability on real data

Load a real wage dataset and compute some of the same quantities you computed by hand in A2.

```
data(wage1, package = "wooldridge")
dt <- as.data.table(wage1)

nrow(dt)      # how many workers?
names(dt)     # what variables?
str(dt)       # types and a peek at the values
```

- a. How many workers are in the data? Name three variables.
- b. Compute the proportion of workers who are female. In the notation of the morning, this is an estimate of $\Pr(\text{female} = 1)$.
- c. Compute the proportion of married workers *among women*, and then the proportion of married workers *among men*. These are estimates of $\Pr(\text{married} \mid \text{female})$ and $\Pr(\text{married} \mid \text{male})$.
- d. Using your answers to (b) and (c), check whether “female” and “married” look independent in this sample. What would independence imply?

Check yourself. You should be able to answer these without notes:

- 1. What is the difference between an outcome, an event, and a random variable?
- 2. Why does $\Pr(A \mid B)$ divide by $\Pr(B)$?
- 3. Why did `c(1, 2, "three")` turn the numbers into text?